THE RADIUS SPECTRUM OF GAMES, AND SIMULTANEOUS *Go* VIA OBJECTIVE QUANTUM REDUCTION

YASHA SAVELYEV

Abstract. Based on the theory of simultaneization developed here, we introduce an isomorphism invariant of combinatorial games valued in subsets of $\mathbb{N} \sqcup \{\infty\}$, called the radius spectrum. It is an invariant of the dynamics, orthogonal to quantitative data such as cardinality, and to the scoring. We then draw a stark line between chess and *Go*: the radius spectrum of *Go* is all of $\mathbb{N} \sqcup \{\infty\}$, of chess $\{0\}$. The proof factors through the construction of a particular simple symmetric simultaneization *SGo* of *Go*, all terms being mathematical notions. *SGo* is striking purely as a game: it is refereed by the display alone, and it is sharper and (much) deeper than *Go* in computer simulations. Its rules are simple, especially mathematically; in fact the Komi, Ko and suicide rules of *Go* are removed. For these reasons we take time to describe it gently and fully. Using the Nash equilibrium theorem, we show that a finite symmetric simultaneous combinatorial game has a mixed equilibrium strategy that draws on average, and so *SGo* does as well. The construction of *SGo* is inspired by physical quantum theory, despite the game being combinatorial.

1. Introduction

At a basic level we study simultaneizations of combinatorial games and use this to derive qualitative invariants of their game dynamics. There are intrinsic reasons why one may want to simultaneize a game – in particular *Go*. The latter is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. But for *Go* this structure creates asymmetries which force departures from local-to-global principles – mainly Ko fights.[1] One idea is that with simultaneization we can remove the Ko rule completely, and of course Komi.

The basic finding is that perhaps due to its local-to-global nature, *Go* is almost simultaneous. We make this precise by proving that *Go* has simple (in a certain mathematical sense) symmetric simultaneizations of faithfulness radius any $k \in \mathbb{N}$ or ∞, see Theorem 5.2. Much of the general theory of simultaneizations is developed in the present paper.

The faithfulness radius calculations lead us to introduce an isomorphism invariant of the dynamics of a game G, orthogonal to quantitative data such as cardinality, and to the scoring – the radius spectrum of G, valued in subsets of $\mathbb{N} \sqcup \{\infty\}$. For *Go* it is $\mathbb{N} \sqcup \{\infty\}$, by Theorem 5.2; in contrast, for chess it is $\{0\}$, by Theorem 5.5.

The proof of Theorem 5.2 factors through the construction of the radius zero game *SGo*, and it is the star here – it is an easy to play game with razor sharp and

[1]If a player engages in a Ko fight, they must make moves that are strategically disconnected.

deep dynamics. It is a combinatorial symmetric simultaneous game, which is played on a standard *Go* board and refereed by the display alone[2], although we need a third type of stone that we call q-stones, represented by colored stones. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules: stone placement and capture (aside from the end of game rule). Like classical *Go*, *SGo* has very robust game dynamics. We make this precise via some empirical studies outlined in Section 2.1 and Section 6.2. In computer simulations random games of *SGo* appear to be sharper than those of *Go*, markedly so at 19×19 (but this must be properly interpreted). More interestingly, *SGo* is a much deeper game than *Go* on 5×5, with an Elo ladder three times the height. These statistics are preliminary; the methodology is coarse and the sample sizes moderate. But *SGo* being deeper should be unsurprising from the construction alone, as the state space is much larger and the evolution rule more intricate.

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal simultaneization of certain sequential games developed in Section 4.4, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics; in the case of *Go* this is seen empirically, cf. Section 2.1. To get interesting dynamics, the *SGo* game evolution (capture rule) invokes "objective reduction of the quantum state" – it is a deterministic, generally easy, objective operation performed on the board. The term objective reduction was coined by Penrose, see for instance [21]; in terms of the branch semantics we are collapsing certain branches.

Of course, the word quantum is used figuratively: there is no unitary time evolution, no complex amplitudes, and no probabilistic outcomes. The game is combinatorial and deterministic, and it would not be appropriate to call it "quantum Go". For games with genuine quantum features see for instance [7], and for a quantum version of *Go* itself [24].

Using the Nash equilibrium theorem [19], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said, finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw, not just to draw on average; on the 2×2 board both players do, but already on the 3×3 board neither player has one, so optimal play is necessarily mixed (Section 4.2).

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [2], [26]. It also led to Conway's theory of surreal numbers [4]. There is also research on simultaneous and synchronized combinatorial games, see for instance [3], [11], [12]. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

2. Game rules of *SGo*

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection

[2]Note that this is beyond even chess and *Go*: the latter have, for instance, castling rights and Ko rules that are not purely display driven.

on the board, but in *SGo* they do this on the same turn without knowledge of the other's move on this turn. We allow pass moves as in *Go*. The board the players see and play on we call the ***display***. Every stone of the display carries a ***time stamp*** – the turn of its placement – as part of the display's data. The capture conditions below read the stamps. Simultaneous action requires modifying the placement and capture rules of *Go*, while removing the Komi, Ko, and suicide rules. We also need to introduce a third type of stone, called ***q-stones*** (quantum stones), which may be thought of as simultaneously white and black. On the board a q-stone is displayed *red* or, as a visual aid, *blue* when a capture is currently blocked through it, in the sense of the capture conditions below; the colors red and blue are purely visual and carry no rule content.

2.1. **Motivation for the rules.** Before we proceed to the main rules, we briefly overview the motivation and constraints we impose. We require:

(1) *SGo* forms a simple simultaneization of *Go*, in the precise sense of Theorem 5.2. Being a simultaneization means that *SGo* naturally maps to the universal simultaneization of *Go*.
(2) It is played on a standard *Go* board and refereed purely by the display. This in particular requires us to represent simultaneous moves to the same intersection, and we choose to do this by introducing q-stones (visually red or blue).
(3) No Ko or other non-repetition rules.
(4) We want *SGo* to have comparable game dynamics to the standard *Go*. This is far from the case for the universal simultaneization Sim(*Go*) of *Go*, as considered in Section 4.4. For Sim(*Go*), games tend to be drawn due to players lacking forcing options.

    To achieve these dynamics, we utilize an "objective reduction" process acting on the display each turn (Section 2.5). This is entirely display driven and elementary. In the language of the universal simultaneization, it prunes certain branched histories resulting from simultaneous action, using objective global board information. This has the effect of giving players very forcing options, see Section 3.2 for some examples. But this is not without constraints: we need the game to stay symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 2.4. We also cannot prune too much, for example in the suicidal capture condition in Definition 2.2 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 2.3.

Requirement (4) is supported by computer experiments.[3] The main comparison plays both games on their common rule base – the variant of *Go* fixed before Theorem 5.2 (suicide legal, no superko rule) – under uniformly random play over all legal board moves, passing only when none is available; 1000 games per variant (200 in the last column). On the 5×5 board the game bound of Section 4.2 is set to 40 turns; on 19×19, *SGo* is left unbounded, while *Go* is stopped at 1500 turns:

[3]The simulations and exact computations reported in this paper, together with extensive machine verification of its rules, examples, and arguments – including an exhaustive census of the one-turn faithfulness conditions over every legal classical state of the 3×3 board – were carried out by Claude, an AI system by Anthropic.

	SGo 5×5	*Go* 5×5	*SGo* 19×19	*Go* 19×19
mean margin	13.6	12.0	235.3	68.8
draws	4.6%	5.4%	0.0%	4.5%
games ending on their own	99%	0%	100%	50%
mean length in turns	22.4	39.9	398	1106
q-stones created per game	3.3	–	8.6	–
q-stones on the final display	2.3	–	5.0	–
peak recursion depth	1.1	–	1.0	–

Here the mean margin is the absolute difference of the two area counts, and a game ends on its own when the end of the game rule fires before the bound or cap (the 19×19 figures for *Go* are censored at the cap, so its true mean length is larger still). On 5×5 the margins are nearly identical and the draw rates similar, and the difference is termination: 99% of *SGo* games ended of their own accord within the bound – and with the bound lifted, every one of them does, in 22.4 turns on average – while every classical game was still churning when the bound stopped it. On 19×19 the difference is dramatic on both counts: random *SGo* ends by itself on a decisively settled board, with mean margin exceeding half the board, while half of the classical games were still cycling, far from settled, when stopped (every one of the thousand *SGo* games ended on its own).

Disabling the objective reduction – approximating the dynamics of the universal simultaneization of Section 4.4 – collapses the game completely: no stone is then ever removed, the two colors necessarily place equal numbers of stones, and all 500 simulated 5×5 games ended in exact draws. The objective reduction is the entire engine of the dynamics.

One may wonder how informed play changes the above distributions; and also what the depth estimates look like; we partially address this in Sections 6.1 and 6.3.

Finally, the last three rows of the table concern the quantum layer. A finished *SGo* game displays on average about two q-stones on the small board, and five among the 361 points of the large one, so the display stays nearly classical. The recursion depth of a turn is the number of capture reductions (Section 2.5) in its resolution that remove a stone. Its peak over a game averages barely above one, and it never exceeded three in the thousand games at either size. In other words, nearly every turn resolves analogously to *Go*, despite the "quantum engine".

2.2. **Some definitions.** Given a board state s, that is any (at the moment no legality restrictions) collection of white, black, and q-stones on the board, each stone implicitly time stamped, let $G(s)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) The display is a board state. From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

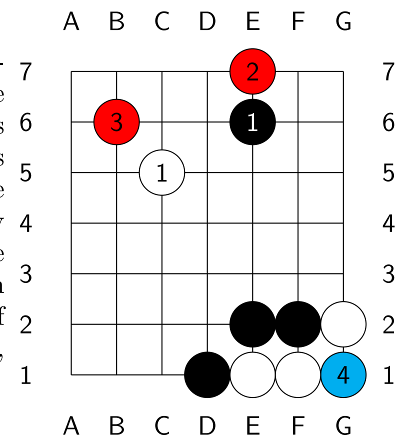
A ***unicolor component*** C is a connected component of the full (a.k.a. induced) subgraph of $G(s)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C ***black***, respectively ***white***. We may less often refer to ***q-components***, analogously defined as connected components of q-stones.

As in classical *Go*, a connected component C ***has no liberties*** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a ***trapped component***.

We define stones ***adjacent*** to C as those stones whose corresponding vertices are connected by an edge to C in $G(s)$.

2.3. **Stone placement rule of *SGo*.** If the players move to the same intersection i on the board, a q-stone is placed at i.

To the right is an example of the placement rule. On turn one the players move to different intersections and both stones are placed as in classical *Go*. On turns two, three and four they move to the same intersection – E7, B6 and G1 respectively – and each conflict yields a q-stone. (One can check that no capture then occurs on turn 4: the blue stone blocks both parts of condition 3 for the white component E1, F1.)



2.4. **Capture rule of *SGo*.** After both White and Black move on the current turn numbered n, using the stone placement rule above we get a display D.

Definition 2.1. We say that a black, white, or q-component C is ***trapped on turn*** n if C is trapped and n is the largest stamp among the stones of C and the stones adjacent to C. A stone is called ***trapped*** on turn n, if it is an element of C as above. We say that a q-stone is ***trapped by white respectively black on turn*** n (white, respectively black, being the capturing side), if replacing it by a black respectively white stone it is trapped on turn n. For a black respectively white stone, trapped by white respectively black on turn n will simply mean trapped on turn n.

Definition 2.2. A white component C is resolved as ***captured on this turn*** if the following is satisfied:

(1) C is trapped on this turn.
(2) If C is adjacent to a q-stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 2.4 does not work, and the game evolution will be ill defined.)
(3) One of the following holds:
  (a) No black stone or q-stone adjacent to C is trapped by white on this turn.
  (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then the component of that stone contains a stone that is created on this turn and is not a q-stone.

Remark 2.3. *The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.1. For example, if we remove the part of the condition "is not a q-stone", then Ko type infinite loops will become possible later. If we remove the condition "no stone in C is created on this turn", then the proof of Theorem 5.2 will fail.*

In the case C is captured:

- The stones of C are removed as in *Go*.
- Any q-stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

A stone converted this way is not considered to be placed or created on this turn; it retains the placement turn of the q-stone it replaces.

If C is a black component the process is naturally mirrored. As far as the rules are concerned this process is simultaneous for all captured components, cf. Lemma 2.4.

2.5. **Capture reduction.** We call the above ***capture reduction***. It is iterated until it removes no stone, and the resulting display is denoted $Cap_n(D)$.

Lemma 2.4. *The capture reduction is well defined. Namely, no q-stone is adjacent to components of both colors captured on this turn, so that the displayed conversion of q-stones is unambiguous.*

Proof. Suppose that a q-stone r is adjacent to unicolor components C_0, C_1, both captured on this turn, with C_0 white and C_1 black. As C_1 is captured, it is in particular trapped. So C_0 is adjacent to the q-stone r, which is itself adjacent to the trapped component C_1 of the opposite color, and Condition 2 fails for C_0. This contradicts the assumption that C_0 is captured.

Thus, each q-stone adjacent to some component captured on this turn is adjacent to captured components of only one color, and so its conversion is unambiguous. As the removal of stones is likewise unambiguous, each simultaneous resolution step, and hence the capture reduction, is well defined. □

If the capture reduction removes no stones, the turn is resolved. Otherwise, define $R^k(D)$ by:

$$R^1(D) = Cap_n(D).$$
$$R^k(D) = Cap_{n-k+1}(R^{k-1}(D)).$$

That is, at stage k the capture conditions of Definition 2.2 are applied with "this turn" read as turn $n-k+1$. The recursion stops as soon as $R^k(D) = R^{k-1}(D)$, and the turn is then resolved. The total effect is the ***objective reduction*** (or just OR) of the turn:

$$OR(D) = R^k(D).$$

Remark 2.5. *In practice the resolution of most turns is easy to work out mentally. This is actually quantifiable; in the statistics of Section 2.1 the peak recursion depth never exceeded three and was typically one.*

2.6. **End of the game.** An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some black/white stones from the board as dead. (As is common in standard play of *Go*.) This is the main way of ending the game.
- The display is unchanged after each of two consecutive turns. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row.
- The display has no empty intersections.

We then decide the winner using Chinese *Go*-style rules. So the winner is decided by comparing the two area counts, which we now explain.

We say that an empty intersection i on the board ***reaches black respectively white*** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is ***surrounded by black*** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a q-stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

(1) The number of black stones on the board.
(2) The number of empty intersections surrounded by black.

See Section 3.6 for an example.

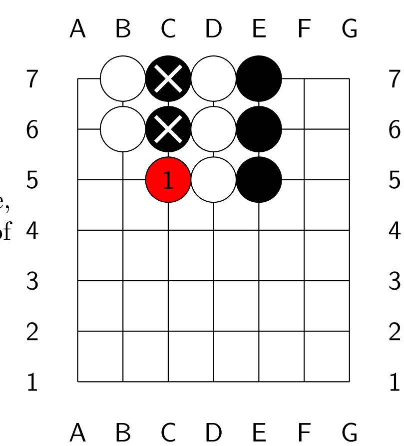
A draw is possible, although compared to chess it is unlikely (for random games on the 19×19 board especially so, as the table in Section 2.1 shows).

3. Examples

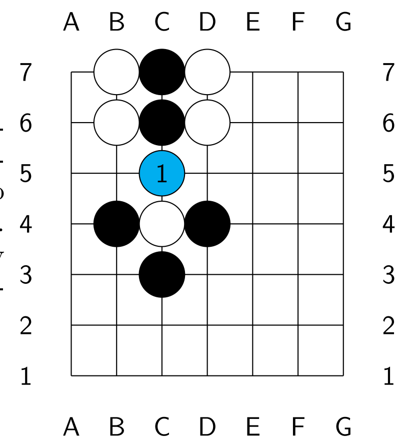
Throughout, a turn's joint move is written m/m', the move of the first player listed first – in *SGo*, Black's.

3.1. **Capture and non-capture moves.** Here are some examples and non-examples of capture.

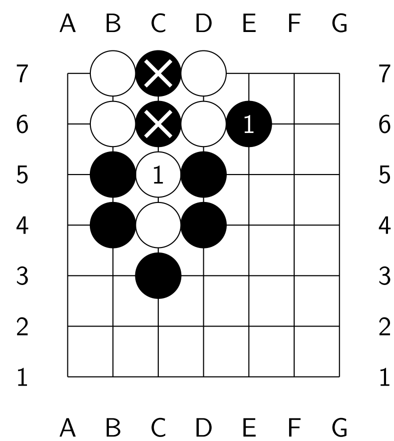
Turn 1 is $C5/C5$. This is a capture move, and C5 will resolve to white at the end of the turn.



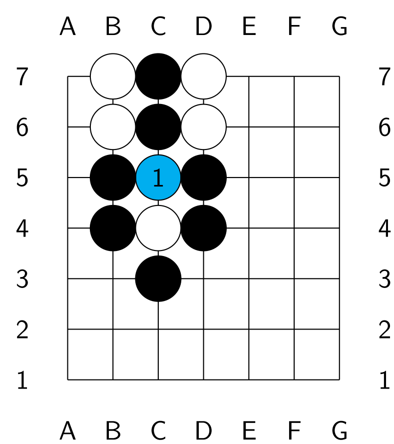
The turn is again $C5/C5$, in another display: not a capture move, as the created q-stone, marked blue, is adjacent to an opposite-colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



Turn 1 is $E6/C5$. White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture.

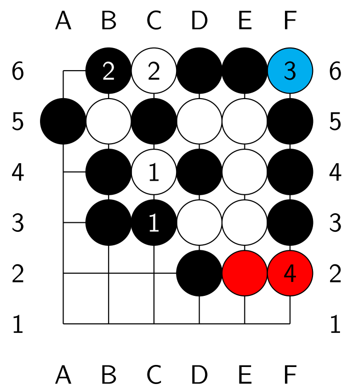


The turn is $C5/C5$: not a capture move, since Condition 3b is now not satisfied, the stone placed on this turn being a q-stone (marked blue, as the blocker).

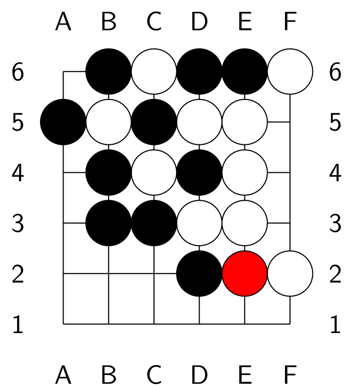


3.2. **More complex examples.**

In the display on the right, turn 1 is $C3/C4$ – Black attempts capture at C3, White at C4. Nothing is captured on turn 1 according to our rules, and after turn 2 we get a display with captures blocked across turns. The resolution below cascades through the stages. The continuation $B6/C6$; $F6/F6$; $F2/F2$ is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section; F6 is marked blue, being what blocks the capture of D6 and E6 on this turn. But F2 is a capture move for White since condition 3a is satisfied.

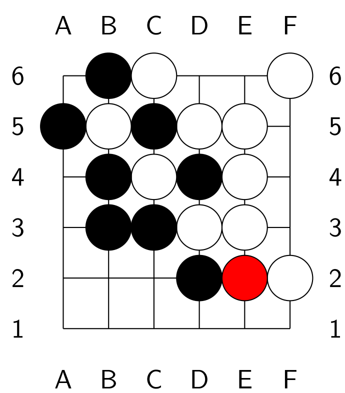


At the end of turn 4, the capture reduction is depicted on the right.



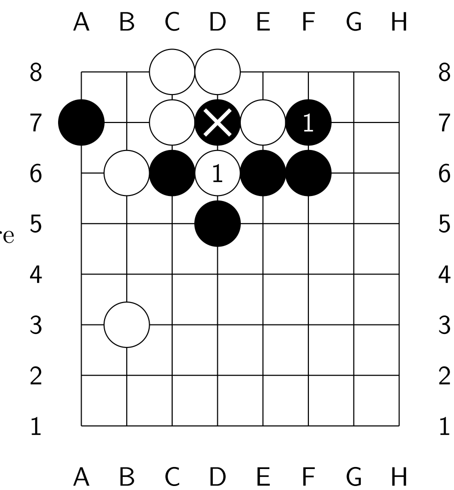
Stage two reduction. Stage three, which applies the capture conditions of turn 2, removes nothing: C5 and B5 each block the other's capture. So the recursion stops, and this is the final resolution of turn 4.

Note that B5, C5, C4 and D4 are all trapped yet uncaptured – an entangled analogue of seki, impossible in classical *Go*: each capture is blocked across turns by the others.

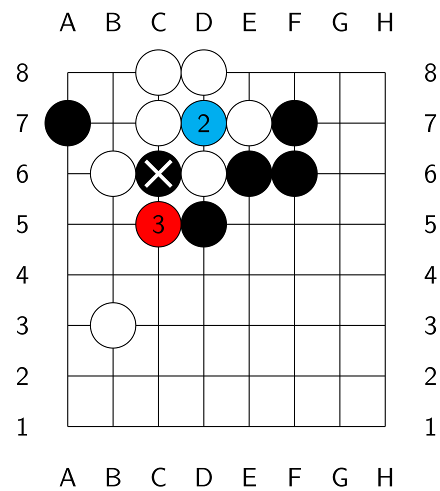


3.2.1. *Ko situation.* Below is a Ko type situation. Recall that we do not have a Ko rule; none is needed, since the display cannot immediately loop by recapture.

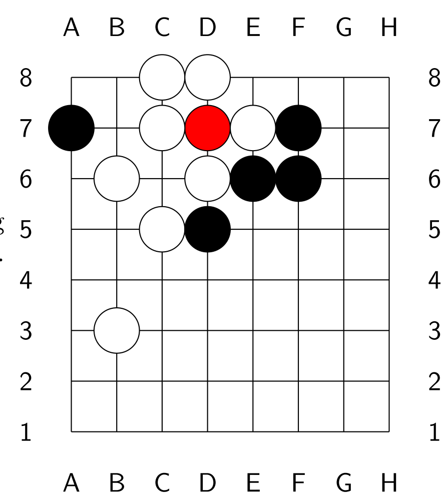
Turn 1 is $F7/D6$; White's D6 is a capture move.



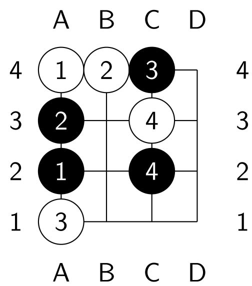
Here is the possible continuation $D7/D7$; $C5/C5$; the blue stone at D7 is what blocks Black's recapture of White's D6.

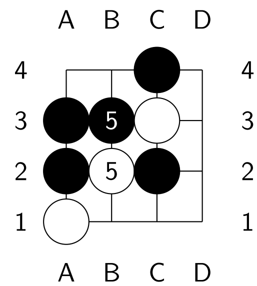


At the end of turn 3 we get the following display: the q-stone at D7 simply persists. The Ko fight remains open.



3.3. **Component clause firing.** Here is an example where the component clause of the suicidal capture condition 3b – that the trapped neighbor's *component* contain a stone created on this turn – does the work. On the 4×4 board play the conflict-free history $A2/A4$; $A3/B4$; $C4/A1$; $C2/C3$:





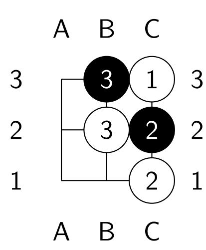
Now play $B3/B2$. Black's B3 fills the last liberty of the old white pair A4, B4, while White's B2 fills the last liberty of the black group A2, A3, B3. The black group is trapped by White on this turn, so condition 3a fails for the pair and the suicidal capture clause governs. The pair touches the trapped group at A3 – an old stone – but the group *contains* this turn's B3: condition 3b is met, and the pair is captured, as the right board shows.

This is what the classical semantics demands: the two serializations of the turn agree, and each captures the pair – B3 played first captures it at once, and B2 then lands with the liberty B1; B2 played first, B3 still captures it, the black group surviving on the freed liberties. Formally, condition 3b is necessitated by condition 4 of Definition 4.16 of simultaneization.

3.4. **Quantum effects.** It helps here to use branch semantics from the formalization of Section 4.1; the rules themselves never consult that formalization, and the reader may ignore it on first reading.

The branches of the entanglement state can interact through the display; we might call these quantum effects.

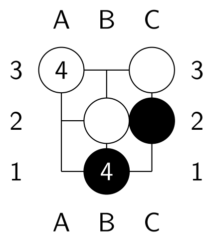
On the 3×3 board play the history pass/$C3$; $C2/C1$; $B3/B2$, arriving at the display:



Both C3 and C2 are trapped yet uncaptured.

In one branch of the entanglement Black's B3 has captured C3; in the other White's B2 has captured C2. This is a standoff on pure black and white stones, with no q-stone anywhere in the history.

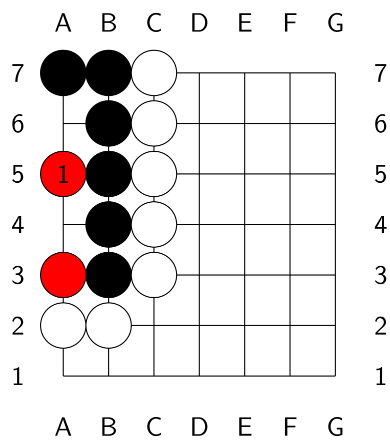
Now play the joint move $B1/A3$. The displayed resolution captures both B3 and C1:



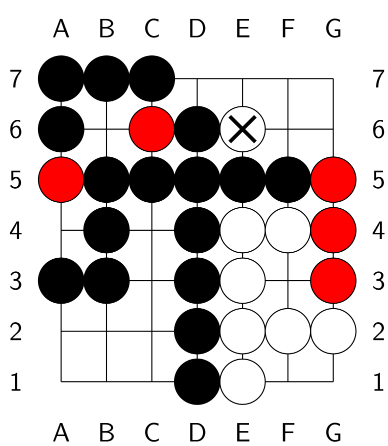
though in each branch only one of them is captured. OR thus performs a joint capture that no serialization of the history performs. In the universal simultaneization of Section 4.4 the branches never interact: its evolution is the union of independent classical evolutions.

3.5. **Life and death.** Life and death is positionally very similar to *Go*; here is an example.

White cannot kill the black group: A4 and A6 are two eyes – a white stone placed in either is immediately captured, while a simultaneous conflict there yields a q-stone. In *SGo* the group is alive outright: even with Black's collaboration – Black filling one eye himself, White taking the other – the q-stones at A3 and A5 block every capture.
The group survives, wrapped in the entanglement.



3.6. **End game scoring.** In the diagram below, the players have agreed to end the game. The white stone at E6 is agreed to be dead, and is removed before counting. White has $7+3$ points and Black has $7\cdot 7-(7+3)-5$ points, where the -5 accounts for the five q-stones, which count for neither player.



4. Symmetric simultaneous combinatorial games

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games. Such formalisms are standard in several communities: games with simultaneous moves appear as the deterministic case of the stochastic games of Shapley [28], and as the concurrent game structures of computer science [1]. Simultaneous play of combinatorial games as such is studied by Huggan–Nowakowski–Ottaway [11], see also the references therein; there, however, the effect of a pair of simultaneous moves must be specified by each ruleset separately, no general procedure being available. The emphasis here, on simultaneizations of a given sequential game and on their simplicity, appears to be new, and in this setting the pseudo pass moves together with the operation # further below do furnish such a general procedure. The inverse passage, representing a sequential game within an intrinsically simultaneous formalism, is standard in general game playing, and is likewise effected by pass-like moves: there the idle player is required to play an explicit "noop" move, cf. [29].

Definition 4.1. A ***combinatorial simultaneous game*** (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the ***pseudo pass move*** (it has multi grading both 0 and 1).
- Game states decompose as
$$S(G) = B(G) \sqcup B(G) \times M(G),$$
where $B(G)$ is some set, its elements the between-turn states.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:
$$E = E_G : S(G) \times M(G) \to S(G),$$
which has the properties:
  1. $E(s, m)$ is undefined for s in one of the final states.
  2. If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
  3. If s is not final, $E(s, 0)$ is defined.
  4. If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' is 0).
  5. Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined, then $E(s, m_0, m_1)$ is defined.
  6. For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and
$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$

We say that G is ***finite*** if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \; E(q_0, m_1, \ldots, m_k) \text{ is defined.}$$

Note that any game history of length $2k$ can be rewritten in the form $h = \{m_i^0, m_i^1\}_{i=1}^{i=k}$, with $m_i^j \in M_j(G)$.

It is our convention that $S(G)$ is the set of legally obtainable states, that is states of the form $E(q_0, m_1, \ldots, m_k)$, for some m_i.

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4.2. We say that a combinatorial simultaneous game G is ***symmetric*** if the following conditions are satisfied.

(1) There is a bijection $\mathcal{A} : M(G) \to M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
(2) There is a bijection $\mathcal{R} : B(G) \to B(G)$, satisfying $\mathcal{R}^2 = id$ and $\mathcal{R}(q_0) = q_0$, which we extend to $\mathcal{R} : S(G) \to S(G)$ by $\mathcal{R}(s, m) = (\mathcal{R}(s), \mathcal{A}(m))$, s.t. for all $s \in S(G)$, all $m \in M(G)$, and for $\mathcal{A}$ as above we have:
   - $\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
   - $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

4.1. **Formalizing** ***SGo*****.** A state s of *SGo* is a pair. The first member is the display, as maintained by the rules of Section 2, its time stamps included. The second is the ***entanglement state*** $\rho(s)$, briefly the *entanglement*: a finite set of classical *Go* states, called ***branches*** – the classical states that the play so far may have produced. Note that this is decorative: it is extra data we choose to baggage for reasons that will be apparent soon. The game rules themselves are completely display driven; the player does not care about this baggage. Later we introduce a variant game *DSGo* whose game evolution does consult the entanglement.

The initial entanglement state is the singleton of the empty state. On a turn, first each branch evolves as a classical *Go* state: the two moves of the turn are played on it in both orders, a move to an occupied intersection forfeited, and the outcomes are collected. The display is resolved by the rules of Section 2, and the resolved display prunes: the new entanglement state consists of the collected branches ***compatible*** with it. Compatible means: every stone of the branch stands on an occupied intersection of the display, and matches the color of the displayed stone there, a q-stone matching either color.

Lemma 4.3. *SGo is an ssc game, that is a symmetric simultaneous combinatorial game.*

Proof. Set $G = SGo$. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pseudo pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is the full data maintained by the rules of Section 2: the display with its time stamps, together with the entanglement state of Section 4.1 – a finite set of legally obtainable states of *Go*. The initial state is the empty display with the entanglement $\{q_0\}$, for q_0 the empty state.

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are moves to empty intersections of the display, then $E((s, m_0), m_1) \in B(G)$ is defined to be the state produced by the rules of Section 2: the display resolves by the placement rule and the objective reduction. Each branch evolves under the serialized moves, and the branches incompatible with the resolved display are discarded.

To see that G is symmetric, let $\mathcal{R} : B(G) \to B(G)$ be the map that interchanges the black and white colors of the stones in every branch (on the display this interchanges black and white and leaves q-stones invariant). And let $\mathcal{A} : M(G) \to M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \to M_1(G)$, which then induces $\mathcal{A}$. Then clearly the conditions of Definition 4.2 will be satisfied. $\square$

Remark 4.4. *We have not formalized $F(SGo)$: ending a game by agreement is not an automaton notion. This caveat is however minor, and is already present in classical Go.*

4.2. **Solvability.** A theorem of Zermelo [27] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly *SGo* cannot have a winning strategy: mirroring a winning strategy via the symmetry operators would yield one for the opponent, and both players cannot win. Furthermore, unless we put an upper bound on the length of a game, or on the number of state repetitions, an *SGo* game can be infinitely long.[4] So we will assume there is such a bound, which will make *SGo* a finite game.

One may ask whether *SGo* has a *strategy to draw*: a pure strategy guaranteeing at least a draw. On the 2×2 board it does, with a caveat: here we drop the game length bound and count an unending game as drawn. In this sense, exact computation shows that both players can force at least a draw. Neither player can force the game to end: the drawing strategies are locks, not finishes. But already on the 3×3 board, with any bound between 6 and 8, neither player has such a strategy: a player following any fixed pure strategy can be made to lose. Optimal play is thus necessarily mixed from the 3×3 board on, and we expect the same on all larger boards – the mixed equilibrium of Theorem 4.5 below is not a formality.

Theorem 4.5. *For any assignment of payoffs to a win, a loss and a draw, a finite combinatorial simultaneous game G (Definition 4.1) has a Nash equilibrium.*

Proof. For a non final $s \in B(G)$ set

$$I_i(s) = \{m \in M_i(G) \,|\, E(s, m) \text{ is defined}\},$$

which is non-empty as it contains the pseudo pass move 0.

A ***pure strategy*** of player P_i assigns to each non final $s \in B(G)$ a move in $I_i(s)$, so that the set of pure strategies is the finite set

$$\Sigma_i = \prod_s I_i(s),$$

[4] This is also true for *Go* and chess.

the product over the non final elements of $B(G)$. A pair of pure strategies determines a play of the game, and so a payoff, by finiteness of G.[5] This presents G in normal form: a finite two-player game with the strategy sets Σ_0, Σ_1. The mixed strategies of player P_i are then $\mathcal{P}_i = \Delta(\Sigma_i)$, the probability distributions over Σ_i. The theorem is now a direct application of the Nash existence theorem [19]. □

Corollary 4.6. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 for a draw. By the normal form presentation in the proof of Theorem 4.5, G is a zero sum matrix payoff game, with the rows and columns of the payoff matrix indexed by the pure strategies Σ_0, Σ_1. By the classical theory of matrix payoff games (see for instance [20]), all Nash equilibria have the same expected payoff for Black, namely the value v of the matrix game, and an equilibrium strategy of Black guarantees Black an expected payoff of at least v against every strategy of White. So it is enough to show that $v = 0$; that a symmetric zero sum game has value zero is classical, cf. [9], and for completeness we give the argument.

Suppose otherwise, and suppose without loss of generality that $v > 0$, so that for an equilibrium joint strategy

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

Black has expected payoff $v > 0$.

Let $\mathcal{R}, \mathcal{A}$ be the symmetry operators as in the proof of Lemma 4.3. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has expected payoff v, that is with Black expected payoff $-v < 0$. As the latter must again be v, we have a contradiction. □

4.3. **Symmetric combinatorial sequential games.** Combinatorial sequential games have a beautiful Conway recursive definition [4]. However, for our purpose, in particular for the universal simultaneization construction, it is convenient to interpret them in terms of finite automata.

Definition 4.7. A ***combinatorial sequential game*** (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the ***pseudo pass move*** (it has multi grading both 0 and 1).
- A $\mathbb{Z}_2$ graded set of game states $S(G) = States(G) = S_0(G) \sqcup S_1(G)$, we let $deg : S(G) \to \mathbb{Z}_2$ denote the associated function.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S_0(G)$, understood as the state in which the game is initialized.

[5] Possibly the finiteness condition can be relaxed.

- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \to S(G),$$

  which has the properties:
  (1) $E(s, m)$ is undefined for s in one of the final states.
  (2) $E(s, m)$ is defined only if m has the same grading as s or if $m = 0$.
  (3) If $E(s, m)$ is defined then $deg(E(s, m)) = deg(s) + 1$.
  (4) If s is not a final state then $E(s, 0)$ is defined.
  (5) If $E(s, 0) = E(s', 0)$ then $s = s'$.
  (6) If $E(s, 0, 0)$ is defined then it is a final state: two consecutive pseudo pass moves end the game.

We recursively define:

$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$

As before it will be understood that $S(G)$ is the set of legally obtainable states, that is states of the form $E(q_0, m_1, \ldots, m_k)$, for some m_i.

Example 4.8. *Note that Go has a pseudo pass move, but chess does not; however, chess can be augmented with a pseudo pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules, and with the game ending in a tie after two consecutive passes. This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in Zugzwang they may just pass). We will call this* **augmented chess** *denoted by* $\widetilde{chess}$.

Definition 4.9. We say that a combinatorial sequential game G is ***symmetric*** if the following conditions are satisfied.

(1) There is a bijection $\mathcal{A} : M(G) \to M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
(2) There is a bijection $\mathcal{R} : S(G) \to S(G)$, satisfying $\mathcal{R}^2 = id$ and taking $S_0(G)$ to $S_1(G)$ s.t. for all $s \in S(G)$, all $m \in M(G)$ and for $\mathcal{A}$ as above we have:

$$\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m),$$

whenever both sides are defined. Furthermore, if one side is defined then so is the other.
(3) $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Example 4.10. *Of course* $\widetilde{chess}$ *and classical Go are examples of symmetric games. In the case of Go,* $\mathcal{A}$ *interchanges black and white moves and* $\mathcal{R}$ *likewise interchanges the color of the stones on the board and shifts the initiative (it is easy to see that this does result in a legally obtainable state).*

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{A}$, $\mathcal{R}$ implicitly fixed).

4.4. **Universal simultaneization.** Let G be a sequential game (by our standing assumption with a pass move 0 and symmetric). Let $\{m_i\}$ be a sequence of game moves. If some $m_i = 0$ and $m_k \neq 0$ for some $k > i$, then we say that it is an ***interior pass move***. We say that a finite sequence $\{m_i\}$ of moves of G is ***pseudo legal*** if it becomes a game history after adding or removing some interior pass moves from the sequence.

For $a \in S(G)$ if there exists an $a' \in S(G)$ s.t. $E_G(a', 0) = a$ we set $\overline{a} = a'$, otherwise set $\overline{a} = E_G(a, 0)$. Property 5 in Definition 4.7 ensures that it does not matter which a' is chosen when it exists. For $a \in S(G)$ and $m \in M(G)$ set

$$a \# m = \begin{cases} E_G(a, m), & \text{if this is defined} \\ E_G(\overline{a}, m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Let $\sim$ denote the ***pass equivalence***: the equivalence relation on the states of G generated by $s \sim E(s, 0)$, for s and $E(s, 0)$ non final. We call its classes ***pass classes***. By Property 5 and Property 6, a pass class contains at most one state of each grading. For a non final $b \in S(G)$ we let $\widehat{b}$, the ***grading normalization*** of b, denote the grading 0 member of the pass class of b when it exists, and $\widehat{b} = E_G(b, 0)$ otherwise; so $\widehat{b} = b$ for $b \in S_0(G)$, and $\widehat{b}$ is pass equivalent to b outside of the degenerate second case, in which a pass is appended. It is convenient to regard final states as carrying both gradings, as with the pseudo pass move, and for final b we set $\widehat{b} = b$. Define $P(a, m_0, m_1) \in 2^{S_0(G)}$, for $a \in S_0(G)$, by the following prescription in decreasing priority, where each term is understood to be replaced by its grading normalization, and the expression is defined if and only if all the terms are defined.

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1 \# m_0$ (note that if $m_0 = m_1 = 0$ then the two terms coincide, so that $P(a, 0, 0) = a \# 0 \# 0$.)
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S_0(G) \times M_0(G) \times M_1(G) \to 2^{S_0(G)},$$

which we call the ***serialization map***: it evaluates the turn in its available serializations – both orders of play when both are defined, the degenerate cases otherwise – and collects the outcomes.

Definition 4.11. Let G be a sequential game. Its ***universal simultaneization*** is the simultaneous game $\mathrm{Sim}(G) = G'$ satisfying the following.

(1) $B(G') = 2^{S_0(G)}$. (The elements s of $\mathfrak{s} \in B(G')$ will be called ***state branches***.) By the above, an element of $B(G')$ may equivalently be regarded as a finite collection of pass classes, represented by their grading 0 members.
(2) The initial state $q_0' = \{q_0\}$.
(3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
(4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$; then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$, s is not final and $s \# m_i$ is defined.
(5) For $\mathfrak{s}$ and m_i as just above:

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} P(s, m_0, m_1),$$

(undefined if some $P(s, m_0, m_1)$ is undefined).
(6)

$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s}\, s \in F(G).$$

Note that $\mathrm{Sim}(G)$ is uniquely defined by the above conditions up to the partition of its final states into W_0, W_1, D. The partition plays no role beyond the symmetry of Lemma 4.13; one may take $\mathfrak{s}$ winning for a player exactly when some branch wins for them and no branch wins for the opponent, and drawn otherwise.

We do not impose the obtainability convention on $\mathrm{Sim}(G)$: $B(\mathrm{Sim}(G))$ is the entire power set. In particular the empty set is a state, non final and with every move available, by the conditions above. It is unreachable within $\mathrm{Sim}(G)$ itself; the branch semantics of Definition 4.16 may reach it.

Remark 4.12. *The passage from states to sets of states is a form of the classical subset construction of automata theory* [23]*. In game theoretic terms, a state of* $\mathrm{Sim}(G)$ *is a knowledge state: the set of classical states consistent with what the players have observed. Analogous knowledge based constructions are used in the study of games of imperfect information, see for instance* [25], [6].

Lemma 4.13. $\mathrm{Sim}(G)$ *is symmetric.*

Proof. Set $G' = \mathrm{Sim}(G)$, and let $\mathcal{A}, \mathcal{R}$ be the symmetry operators of G. Set $\widetilde{\mathcal{A}} : M(G') \to M(G')$ to just be $\mathcal{A}$. For

$$\{s_0, \ldots, s_k\} = \mathfrak{s} \in B(G'),$$

set

$$\widetilde{\mathcal{R}}(\mathfrak{s}) = \{\mathcal{R}(s_0), \ldots, \mathcal{R}(s_k)\},$$

with each $\mathcal{R}(s_i)$ replaced by its grading normalization, $\mathcal{R}$ being grading reversing. Since $\mathcal{A}(0) = 0$, the map $\mathcal{R}$ commutes with the pass move and so preserves pass equivalence; with Property 6 it follows that $\widetilde{\mathcal{R}}^2 = id$ exactly, and that the computation below is unaffected by the normalization. For

$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$

set

$$\widetilde{\mathcal{R}}(\mathfrak{s}, m) = (\widetilde{\mathcal{R}}(\mathfrak{s}), \widetilde{\mathcal{A}}(m)).$$

The needed symmetry properties for the maps $\widetilde{\mathcal{A}}, \widetilde{\mathcal{R}}$ are then verified using the symmetry properties of $\mathcal{A}, \mathcal{R}$ as follows. It is easily seen that $\overline{\mathcal{R}(a)} = \mathcal{R}(\overline{a})$, hence

$$\mathcal{R}(a)\#\mathcal{A}(m) = \begin{cases} E_G(\mathcal{R}(a), \mathcal{A}(m)) = \mathcal{R} \circ E_G(a, m), & \text{if defined} \\ E_G(\mathcal{R}(\overline{a}), \mathcal{A}(m)) = \mathcal{R} \circ E_G(\overline{a}, m), & \text{if defined} \end{cases}$$

and so

$$\mathcal{R}(a)\#\mathcal{A}(m) = \mathcal{R}(a\#m), \tag{4.14}$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$; then we get:

$$\begin{aligned} \widetilde{\mathcal{R}}(E_{G'}(\widetilde{\mathcal{R}}(\mathfrak{s}), \widetilde{\mathcal{A}}(m_0), \widetilde{\mathcal{A}}(m_1))) &= \widetilde{\mathcal{R}} \circ \cup_{s\in\mathfrak{s}} P(\mathcal{R}(s), \mathcal{A}(m_0), \mathcal{A}(m_1)) \\ &= \widetilde{\mathcal{R}} \circ \cup_{s\in\mathfrak{s}} \mathcal{R} \circ P(s, m_0, m_1) \text{ using (4.14)} \\ &= \cup_{s\in\mathfrak{s}} P(s, m_0, m_1) \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies property 2 of Definition 4.2 for pair moves at states of $B(G')$; the rest is similar. ☐

Definition 4.15. Let G_0 be a sequential game and G_1 a symmetric simultaneous game with $M(G_1) = M(G_0)$. The ***semi-classical*** states of G_1, with their ***associated states*** in G_0, are defined recursively: the initial state of G_1 is semi-classical, with associated state the initial state of G_0. If s is semi-classical with associated state a, and m_0, m_1 *commute at* a – both $a\#m_0\#m_1$ and $a\#m_1\#m_0$ are defined and pass equivalent – then $E_{G_1}(s, m_0, m_1)$, if defined, is semi-classical, with associated state the common grading normalization of $a\#m_0\#m_1$ and $a\#m_1\#m_0$. A history of commuting turns witnessing this recursion we call a *generating history* of the state it reaches.

Definition 4.16. Let G_0 be a sequential game. We say that G_1 is a ***simultaneization of*** G_0 if the following is satisfied.

(1) G_1 is a symmetric simultaneous game.
(2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
(3) There is a map $\rho : B(G_1) \to B(\mathrm{Sim}(G_0))$ – the *branch semantics* – satisfying $\rho(q_0) = q_0'$ and s.t. for $s \in B(G_1)$: whenever the state $E_{G_1}(s, m_0, m_1)$ is defined, so is $E_{\mathrm{Sim}(G_0)}(\rho(s), m_0, m_1)$, and
$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\mathrm{Sim}(G_0)}(\rho(s), m_0, m_1).$$
(4) For every semi-classical s with associated state a (Definition 4.15): $\rho(s) = \{a\}$.

For $a \in S(G_0)$ set
$$\mathbb{I}(a) = \{m \in M(G_0) \,|\, a\#m \text{ is defined}\},$$
which we call the ***interface*** of the state a: the set of moves of either player available at a, with the operation $\#$ absorbing the turn order. Likewise, for $s \in B(G_1)$ set $I(s) = \{m \in M(G_1) \,|\, E_{G_1}(s, m) \text{ is defined}\}$, the interface of the state s.

Lemma 4.17. *Let G_1 be a simultaneization of G_0. Then for every non final $s \in B(G_1)$: every branch of $\rho(s)$ is non final, and*
$$I(s) \subseteq \bigcap_{a \in \rho(s)} \mathbb{I}(a).$$
That is, a move available at s is available in every branch. Equality of the two sides, together with $\rho(s) \neq \emptyset$, is precisely the statement that a joint move is available at s if and only if it is available in $\mathrm{Sim}(G_0)$ *at $\rho(s)$.*

Proof. First, for $m \in M(G_1)$, $E_{G_1}(s, m)$ is defined if and only if the pair move $E_{G_1}(s, m, 0)$ (in the order of the pair prescribed by the grading of m) is defined. Indeed, the latter presupposes the former, and conversely if $E_{G_1}(s, m)$ is defined then so is $E_{G_1}(s, 0)$, as s is not final, and so by the properties of Definition 4.1 the pair move is defined, using that the pseudo pass move has both gradings. If some $a \in \rho(s)$ were final, then no joint move would be defined in $\mathrm{Sim}(G_0)$ at $\rho(s)$, by the construction of $\mathrm{Sim}(G_0)$; by condition 3 no joint move would then be defined at s, contradicting the definedness of the pair $(0, 0)$ at the non final s. So every branch is non final. Now if $m \in I(s)$, the pair move is defined at s, hence, by condition 3, defined in $\mathrm{Sim}(G_0)$ at $\rho(s)$, which says exactly that $a\#m$ and $a\#0$ are defined for every $a \in \rho(s)$, that is $m \in \mathbb{I}(a)$ for every a. The equality statement is the same unwinding read in both directions. □

Definition 4.18. Let G_1 be a simultaneization of G_0. We say that G_1 is ***faithful*** at a state $s \in B(G_1)$ if $\rho(s)$ is nonempty, s is final only if some branch of $\rho(s)$ is final, and, when s is not final,

$$I(s) = \bigcap_{a \in \rho(s)} \mathbb{I}(a),$$

that is, exactly the joint moves of $\mathrm{Sim}(G_0)$ at $\rho(s)$ are available at s. The finality clause says that G_1 may not invent endings where it is faithful. For $d \geq 0$, we say that G_1 is ***faithful within distance*** d ***of classical play*** if it is faithful at every state reachable from a semi-classical state in at most d turns, distance 0 being the semi-classical states themselves. The ***faithfulness radius*** of G_1 is the supremum of the d such that G_1 is faithful within distance d of classical play, possibly 0 or ∞: the depth to which its branch semantics stays exact. It is defined when G_1 is faithful within distance 0; otherwise G_1 has no radius.

Definition 4.19. Let G_0 be a sequential game. We will say that a simultaneization G_1 of G_0 is a ***simple simultaneization of*** G_0 if for every non final $s \in B(G_1)$ at which a non-pass move is available there is a legally obtainable state $a \in S(G_0)$ with

$$\mathbb{I}(a) = I(s),$$

where for s the initial state of G_1, a can be taken to be q_0. The ***radius spectrum*** of G_0 is the set of faithfulness radii of its simple simultaneizations, a subset of $\mathbb{N} \sqcup \{\infty\}$; a simultaneization with no radius contributes nothing. The ***absolute radius*** of G_0 is the supremum of its radius spectrum.

The radii and the radius spectrum are dynamical isomorphism invariants in the sense of the following.

Lemma 4.20. *Let G_0, G_0' be sequential games, and let φ be a bijection of their states and of their graded moves intertwining the evolutions and the initial states. Then:*

1. *φ intertwines the pass moves and the final states.*
2. *If H is a simultaneization of G_0, then renaming each move m of H to $\varphi(m)$ yields a simultaneization H' of G_0', isomorphic to H over φ. The faithfulness radii of H and H' coincide, and H' is simple if and only if H is.*
3. *The radius spectra of G_0 and G_0' coincide; in particular so do the absolute radii. The spectrum is independent of the scoring: repartitioning the final states into W_0, W_1, D does not change it.*

Proof. (1) A bijection of graded moves carries $M_0(G_0) \cap M_1(G_0)$ onto $M_0(G_0') \cap M_1(G_0')$; each intersection consists of the pseudo pass move. Finality is encoded in the evolution: a state s is final if and only if $E(s, 0)$ is undefined, by properties 1 and 4 of Definition 4.7. Intertwining the evolutions preserves definedness, so φ intertwines the final states.

(2) By (1), φ intertwines the operation $\#$: it is built from the evolutions, the pass moves and the grading (by Definition 4.7 the state grading is the turn parity from the initial state). Hence φ induces an isomorphism $\mathrm{Sim}(\varphi) : \mathrm{Sim}(G_0) \to \mathrm{Sim}(G_0')$, acting on a branch set elementwise. Let H' be H with each move m renamed to $\varphi(m)$: the same states, the evolution and the symmetric structure carried along,

and the branch reduction $\rho' = \mathrm{Sim}(\varphi) \circ \rho$. The conditions of Definition 4.16 for (H', ρ') are those for (H, ρ), transported: the semi-classical recursion of H' relative to G_0' is that of H relative to G_0, with associated states $\varphi(a)$, so $\rho'(s) = \{\varphi(a)\}$ there. The identity on states, with φ on moves, is then an isomorphism over φ. Interfaces satisfy $I_{H'}(s) = \varphi(I_H(s))$ and $\mathbb{I}(\varphi(a)) = \varphi(\mathbb{I}(a))$; finality in H' is finality in H; distances of classical play are turn counts. So the faithfulness conditions of Definition 4.18 hold at s in H' if and only if they hold in H, and the radii coincide. For simplicity: a legally obtainable witness a for s in G_0 maps to the legally obtainable witness $\varphi(a)$ in G_0', with $\mathbb{I}(\varphi(a)) = I_{H'}(s)$; symmetrically for φ^{-1}. Nothing above reads the partition of the final states into wins and draws.

(3) Renaming along φ and along φ^{-1} are mutually inverse, so the simple simultaneizations of G_0 and of G_0' correspond, with equal faithfulness radii; the spectra coincide, and with them the suprema. For the scoring statement apply the lemma to the identity bijection. □

5. Main theorems

5.1. **The decoherent variant *DSGo*.** We now introduce a theoretical variant *DSGo*[6]: *SGo* with the objective reduction preempted by decoherence. Each turn, the stones are placed by the placement rule of *SGo*, and nothing else is played out on the display. The resulting state is then postprocessed as follows. The display is measured against the branch evolution of the turn; the entanglement does the measuring and the removing. A state of *DSGo* carries the same data as one of *SGo* – a display together with its entanglement state $\rho(s)$ – and moves are available as in *SGo*.

The serialization map extends to sets of classical states elementwise,

$$P(\mathfrak{B}, m_0, m_1) = \bigcup_{a \in \mathfrak{B}} P(a, m_0, m_1);$$

we call $P(\rho(s), m_0, m_1)$ the *decoherence* of the turn m_0/m_1 at s.

For an intersection i and a classical state b, let $ev_i(b) \subseteq \{\text{black}, \text{white}\}$ be the color of the stone of b at i, the empty set if i is empty in b. Extend ev_i to a set of classical states by union, and to a display, or a state of *DSGo*, by reading the display, a q-stone contributing $\{\text{black}, \text{white}\}$.

Next, the *postprocessing* $\Delta(s, \mathfrak{M})$ of a state s of *DSGo* by a finite set $\mathfrak{M}$ of classical states, anchored at the current turn, is defined as follows. Set D_0 the display of s and $\mathfrak{M}_0 = \mathfrak{M}$. Proceed recursively; for $j \geq 0$, postprocessing round j removes from D_j the stones at the intersections i with $ev_i(\mathfrak{M}_j) = \emptyset$. A q-stone of D_j created on this turn, at an intersection i with $ev_i(\mathfrak{M}_j) = \{c\}$, becomes a c stone. This gives D_{j+1}.

Based on D_{j+1} the entanglement is then re-cut:

$$\mathfrak{M}_{j+1} = \{b \in \mathfrak{M}_j \,|\, \forall i : \ ev_i(D_{j+1}) = \emptyset \implies ev_i(b) = \emptyset\}.$$

The recursion stops at the first j with $D_{j+1} = D_j$; such a j exists as infinite descent is impossible.

Then $\Delta(s, \mathfrak{M})$ has display D_{j+1} and entanglement state $\mathfrak{M}_{j+1}$. The evolution of *DSGo* is the postprocessed placement:

$$E_{DSGo}(s, m_0, m_1) = \Delta\big(\mathrm{Pl}(s, m_0, m_1),\ P(\rho(s), m_0, m_1)\big),$$

[6] D stands for decoherence. Not meant to be humanly playable without computer resolution.

where $\mathrm{Pl}(s, m_0, m_1)$ is s with the turn's stones placed by the placement rule of SGo and no objective reduction applied. The preemption is forced. Were the objective reduction to act first, its captures standing, every element of the decoherence could be lost in the re-cut; faithfulness would then fail at distance two.

As the proof of Theorem 5.2 shows, the postprocessing loses no element of the decoherence. Hence $\rho \circ E_{DSGo} = E_{\mathrm{Sim}(Go)} \circ \rho$: the entanglement component of a $DSGo$ history is a $\mathrm{Sim}(Go)$ history. $DSGo$ referees $\mathrm{Sim}(Go)$ on a display, differing from it only in scoring: the winner is decided as in SGo, Section 2.6, from the display.

The construction interpolates. For $k \geq 1$, the delayed game k-$DSGo$ stays the execution of the objective reduction for k turns. Formally, the objective reduction $D \mapsto OR(D)$ of Section 2.5, anchored at the current turn, only removes stones and converts q-stones. The *verdict* $\mathrm{ver}(D)$ is the difference: the set of stones of D removed in $OR(D)$, and the set of q-stones of D made definite in $OR(D)$, with their acquired colors. For a verdict v, the execution $v \cdot D'$ removes and recolors accordingly the stones of D' that v names, a stone being named by its intersection and its time stamp; named stones no longer standing in D' are ignored.

A state of k-$DSGo$ is a tuple $(D, \mathfrak{M}, v_1, \ldots, v_k)$: a display, an entanglement state, and the verdicts of the last k turns, v_1 the oldest. The initial state is that of $DSGo$ with all k verdicts empty. Moves, scoring, and finality are those of $DSGo$, read from D and $\mathfrak{M}$. The evolution is

$$E_k\big((D, \mathfrak{M}, v_1, \ldots, v_k), m_0, m_1\big) = \big(D', \mathfrak{M}', v_2, \ldots, v_k, \mathrm{ver}(D')\big),$$

$$(D', \mathfrak{M}') = \Delta\big(\mathrm{Pl}((v_1 \cdot D, \mathfrak{M}), m_0, m_1),\ P(\mathfrak{M}, m_0, m_1)\big):$$

the oldest verdict is executed, the turn is then a turn of $DSGo$, and the new verdict is stored. Verdicts still pending when the game ends are discarded: the display is scored as it stands. As $k \to \infty$ no verdict ever executes, and the game is $DSGo$; SGo, which plays its OR at once and carries no decoherence, stands at the origin of the hierarchy.

Lemma 5.1. *Let s be a state of $DSGo$ and $\mathfrak{M}$ a finite set of classical states.*

(1) *If $b_0 \in \mathfrak{M}$ satisfies, for every intersection i, $ev_i(s) = \emptyset \implies ev_i(b_0) = \emptyset$, then $b_0 \in \rho(\Delta(s, \mathfrak{M}))$.*
(2) *For every intersection i: $ev_i(\Delta(s, \mathfrak{M})) = \emptyset \iff ev_i(\rho(\Delta(s, \mathfrak{M}))) = \emptyset$.*

Proof. In the rounds $(D_j, \mathfrak{M}_j)$ defining $\Delta(s, \mathfrak{M})$ – the recoloring does not change occupancy and plays no role – for (1) we show by induction on j: $b_0 \in \mathfrak{M}_j$, and $ev_i(D_j) = \emptyset \implies ev_i(b_0) = \emptyset$ for all i. For $j = 0$ this is the hypothesis. For the step: an intersection emptied by round $j+1$ has $ev_i(\mathfrak{M}_j) = \emptyset$, and $ev_i(b_0) \subseteq ev_i(\mathfrak{M}_j)$ since $b_0 \in \mathfrak{M}_j$. With the hypothesis, $ev_i(D_{j+1}) = \emptyset \implies ev_i(b_0) = \emptyset$, hence $b_0 \in \mathfrak{M}_{j+1}$.

For (2): let D and $\mathfrak{M}_*$ be the final display and entanglement, so $\mathfrak{M}_* = \rho(\Delta(s, \mathfrak{M}))$. If $ev_i(D) = \emptyset$ then $ev_i(b) = \emptyset$ for every $b \in \mathfrak{M}_*$, by the re-cut; so $ev_i(\mathfrak{M}_*) = \emptyset$. Conversely, a stone of D at an intersection i with $ev_i(\mathfrak{M}_*) = \emptyset$ would be removed by a further round, and the repetition has stabilized. ☐

Let Go be the classical game with Chinese style area scoring and standard pass move, with the following stipulations: suicide is a legal move, as for example in the Tromp–Taylor rules [31, 30], there is no Ko or superko rule, and the game is ended analogously to the way *SGo* is ended, so that the role of a superko rule is played by the end of the game rules.

Theorem 5.2. *SGo, the delayed games k-DSGo for $k \geq 1$, DSGo, and* $\mathrm{Sim}(Go)$ *are simple simultaneizations of Go. Their faithfulness radii are, respectively, zero, exactly k, infinity, and infinity. In particular the radius spectrum of Go is all of* $\mathbb{N} \sqcup \{\infty\}$.

Proof. Call *SGo*, the delayed games *k-DSGo*, and *DSGo* the *SGo family*. Conditions 1 and 2, for the family, are by construction and a direct analogue of Lemma 4.3.

Let $\rho : B(SGo) \to B(\mathrm{Sim}(Go))$ send a state to its entanglement component: the entanglement state is a finite set of states of *Go*, that is an element of $B(\mathrm{Sim}(Go))$ (states of *Go* being identified with their pass classes, via the grading normalization in the definition of P). Then $\rho(q_0) = \{q_0\} = q_0'$. For *DSGo* and the delayed games, ρ likewise reads the entanglement component $\mathfrak{M}$. The branch evolution of Section 4.1 is precisely the evolution by the serialization map P: playing the two moves in both orders, with a move to an occupied intersection forfeited, is the prescription defining P. So $E_{SGo}(s, m_0, m_1)$ is obtained by applying the *OR* prunings to $E_{\mathrm{Sim}(Go)}(\rho(s), m_0, m_1)$, and the inclusion in condition 3 holds by construction.

For the definedness direction of condition 3: a move of *SGo* is available if and only if its intersection is empty on the display; a branch kept by the reduction has all its stones within the display, a branch occupying a display-empty intersection being incompatible with the display and discarded; so a display-empty intersection is empty in every branch, and the joint move is available in $\mathrm{Sim}(Go)$ at $\rho(s)$. (For the rest of the family this is the re-cut, verified below.)

For condition 4, we need: along a generating history of Definition 4.15, the entanglement remains the singleton of the associated state. Fix a generating history h. Write s for a state along h, $\hat{a}$ for its associated state, and $\hat{a}'$ for the associated state after the turn m_0/m_1 of h played at s. The turn commutes at $\hat{a}$, so

$$P(\{\hat{a}\}, m_0, m_1) = \{\hat{a}'\} :$$

the branch evolution carries the singleton to the singleton. The content is the display. The *diagram* of a classical state is its board picture; we say the display is the diagram when the two agree with stamps forgotten. For *DSGo* the verification is an induction along h: the display of s is the diagram of $\hat{a}$, and $\rho(s) = \{\hat{a}\}$. For the step:

$$E_{DSGo}(s, m_0, m_1) = \Delta\big(\mathrm{Pl}(s, m_0, m_1),\ \{\hat{a}'\}\big).$$

The placements match those of $\hat{a}'$. The rounds remove exactly the stones $\hat{a}'$ lacks, and $\hat{a}'$ survives the re-cut, by Lemma 5.1(1). So the new display is the diagram of $\hat{a}'$, and the new entanglement is $\{\hat{a}'\}$.

For *k-DSGo* one line is added: the display of a turn of h is the diagram of a legal state, every component of which has a liberty, so the *OR* captures nothing and the stored verdict is empty; the executions change nothing, and the proof above applies.

For *SGo*: by induction along h the display of s is the diagram of $\hat{a}$, and $\rho(s) = \{\hat{a}\}$. At the turn the two serializations agree, and Lemma 5.3 below resolves the display to the diagram of $\hat{a}'$.[7] The pruning keeps $\hat{a}'$: ρ of the new state is $\{\hat{a}'\}$.

[7] The commuting hypothesis of the lemma is essential. In the first example of Section 3.1 the *OR* performs the captures of one serialization and discards the other. In a mutual capture race,

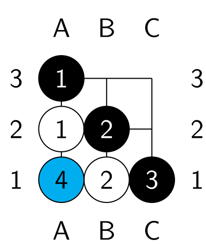
To verify the simplicity property: since in our version of *Go* suicide is a legal move and there is no Ko rule, for any non final state a of *Go* the interface $\mathbb{I}(a)$ consists of the pass moves and the moves of either player to the empty intersections of a. Likewise, for a non final $s \in B(SGo)$, $I(s)$ consists of the pass moves and the moves of either player to the empty intersections of s. So it suffices to exhibit a legally obtainable state $\phi(s)$ of *Go* with the same empty intersections as s. Let $\phi(s)$ be obtained by replacing every stone of s by a black stone. It is easy to see that $\phi(s)$ is legally obtainable.

The simplicity argument applies verbatim to the rest of the family. For these games, the inclusion of condition 3 is the re-cut of Δ, by definition; and a surviving element occupies only display-occupied intersections, which gives the definedness direction as before.

For Sim(*Go*), take ρ the identity. It is a symmetric simultaneous game with the moves of *Go*: Definition 4.11 and Lemma 4.13. Condition 3 is an equality. Condition 4 follows from the recursion of Definition 4.15: a commuting pair evolves a singleton to a singleton. By construction, a move is available at a state $\mathfrak{s}$ if and only if it is available in every branch: $I(\mathfrak{s}) = \bigcap_{b \in \mathfrak{s}} \mathbb{I}(b)$. Also by construction, $\mathfrak{s}$ is final if and only if some branch is final. So Sim(*Go*) is faithful at every reachable state, and its faithfulness radius is infinite. For its simplicity: when a non-pass move is available at $\mathfrak{s}$, some intersection is empty in every branch. The monochrome state with a black stone on every other intersection has the interface $I(\mathfrak{s})$, and it is legally obtainable. So Sim(*Go*) is simple.

It remains to compute the radii of the family.

For *SGo*: at a semi-classical state s with associated state a, the display is the diagram of a and $\rho(s) = \{a\}$, as established above; so $I(s) = \mathbb{I}(a)$, and the finality clause holds, the end of the game rules of the two games being matched by the convention fixed above. Thus *SGo* is faithful within distance zero. It is not faithful within distance one, on any board: reach, by commuting turns, black stones on A3, B2, C1 and white stones on A2, B1, and play the conflict A1/A1, reaching a state s':



In one serialization Black's A1 captures A2 and B1; in the other White's A1 is a suicide, and Black's A1 then lands on the emptied intersection. Both serializations produce the same state, with A2 and B1 empty; the entanglement of s' is its singleton. On the display, the created q-stone is trapped by White on this turn, so Condition 3a fails for each white stone; and Condition 3b fails on its component clause, the q-stone's component carrying no stone of the turn that is not a q-stone.

where each serialization captures a side, the *OR* captures neither, by design. At entangled states the example of Section 3.4 is built on the failure: each of its two captures is performed in some branch, but no single branch performs both. Every branch is then incompatible with the display, and the entanglement empties.

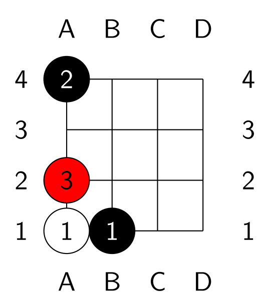
So the OR removes nothing (compare the last example of Section 3.1). So A2 and B1 stand on the display yet are empty in every element of the entanglement: $I(s') \subsetneq \bigcap_{b \in \rho(s')} \mathbb{I}(b)$. The radius of SGo is zero.

For the radius of $DSGo$: we first formalize final states of $DSGo$ and the delayed games as follows. A state s is final exactly when some branch of $\rho(s)$ is final (of course this requires fixing a formalization of final states of Go). Then the finality clause holds by definition.

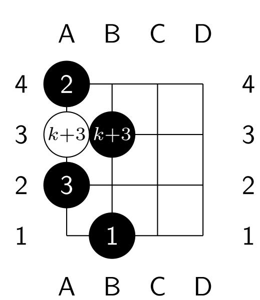
For the move clause, at any reachable state s and any turn, with s' the resulting state: by the re-cut of Δ, every element of $\rho(s)$ occupies only display-occupied intersections. The placement empties no intersection, so every element of the decoherence $P(\rho(s), m_0, m_1)$ has its stones among the display stones and the placements. Each element therefore satisfies the hypothesis of Lemma 5.1(1), and survives: $\rho(s')$ is the entire decoherence, and it is nonempty. By Lemma 5.1(2), an intersection is empty on the display of s' if and only if it is empty in every element of $\rho(s')$, that is $I(s') = \bigcap_{b \in \rho(s')} \mathbb{I}(b)$: faithfulness at s'. So $DSGo$ is faithful at every state reachable in play, and its faithfulness radius is infinite.

For the radius of k-$DSGo$: within distance k of classical play, every verdict that executes was stored at a turn of the generating history, and is empty, as shown above. So every turn within this distance is a turn of $DSGo$, and the preceding argument applies: k-$DSGo$ is faithful within distance k.

It is not faithful within distance $k+1$, on any board. Reach, by commuting turns, a black stone on B1 and a white stone on A1 – with a black stone on A4 on boards beyond 3×3 – and play the conflict A2/A2:



The conflict resolves by the lowest priority case of the serialization map P: the entanglement becomes $\{a_0, a_1\}$, where in a_0 Black's A2 has captured A1, and in a_1 White's A2 has joined A1. The verdict of the new display – A1 removed, the q-stone made definite as black – is stored. After $k-1$ turns of double passes, play B3/A3, reaching a state s' at distance $k+1$ of classical play:



In E_k: the verdict execution has emptied A1 and recolored the q-stone black, its stamp kept; the placement has added B3 and A3. In $P(a_0, \mathrm{B3}, \mathrm{A3})$ the white A3 has no liberty – A4 closes the last – and is gone, a suicide in one order and a capture in the other; write a_0' for the single outcome. Every element of $P(a_1, \mathrm{B3}, \mathrm{A3})$ occupies A1, empty on the display, and the re-cut of Δ removes them: $\rho(s') = \{a_0'\}$. A3 is empty in a_0' yet stands on the display. The verdict execution changes only the display. The elements of $P(a_1, \mathrm{B3}, \mathrm{A3})$ therefore still stand at the removal round of Δ: they occupy A3 and protect it. They are discarded only at the re-cut, and the recursion stops with the display unchanged. So $I(s') \subsetneq \bigcap_{b\in\rho(s')} \mathbb{I}(b)$, and the radius of k-*DSGo* is exactly k.

The proof is now complete. For extra peace of mind on the 3×3 board the theorem is computer verified by exhaustion, setting $k = 1, 2$ for k-*DSGo*. □

Lemma 5.3 (One-stage resolution of commuting turns)**.** *Let $\hat{a}$ be a legal state of Go, D its diagram with all stamps below n, and m_0/m_1 a joint move commuting at $\hat{a}$, with $\hat{a}'$ the common grading normalization. Let D^+ be D with the turn's stones placed. Then the objective reduction of the turn consists of capture reductions at turn n alone, and $OR(D^+)$ is the diagram of $\hat{a}'$.*

Proof. Components, liberties and adjacency refer to D^+ unless a position is named. A commuting joint move places on distinct empty intersections, or passes: a shared intersection forfeits one move in each order, and the outcomes differ by that stone. So D^+ carries no q-stone, and condition 2 and the q-clauses are vacuous. Call a component *fresh* if it contains a stone of the turn, and *old* otherwise. Since $\hat{a}$ is legal and the turn fills only the fresh points, a trapped component is trapped on turn n, and an old component not adjacent to a fresh stone keeps a liberty of $\hat{a}$. Two facts are used throughout. Removals never fill an intersection, and the only fills are the fresh points, which are empty in D; so an intersection emptied during a serialization stays empty to its end. And within the resolution of one placement, the enemy captures run before the suicide check.

Step 1 (removal dichotomy). A group removed at some moment of a serialization is libertyless there; the position at that moment has at most the stones of D^+, so the group is libertyless in D^+. A stone never removed and adjacent to a removed group of its color was present at that group's removal, hence part of it: a contradiction. So in each serialization every component of D^+ is removed entirely or survives entirely, and a removed component is trapped. The two serializations produce the same state, so they remove the same stones. Call a component *dead* if it is removed – in both serializations – and *live* otherwise.

Step 2 (untrapped components are live). An untrapped component has a liberty in D^+; that intersection is never filled, so the component is never libertyless.

Step 3 (old trapped components are dead). Let T be old and trapped, white say. Every liberty of T in $\hat{a}$ is a fresh point; a white fill would merge into T; so T has exactly one liberty in $\hat{a}$, the black point m_0. In the order playing m_0 first no removal precedes, T is libertyless when the enemy captures of m_0 run, and T is captured. By Step 1, T is dead.

Step 4 (live trapped components are fresh, with a dead donor). Let B be trapped and live. At the end of a serialization B has a liberty, so some adjacent component was removed; a same-colored adjacent component is B itself, so the donor is an enemy component, and it is dead. By Step 3, B is fresh.

Step 5 (no two dead components are adjacent). Suppose dead T and dead Z are adjacent, and run either serialization. Both are removed in it. They are not removed at the same resolution: within one resolution the enemy captures of the placed color run first, and the removal of the first of the pair frees the other before its check. So one of them, say Z, is removed strictly earlier. From that moment the spots of Z are empty to the end, and T holds one of them as a liberty. A group holding a liberty is not removed. So T survives the serialization: a contradiction.

Step 6 (the reduction removes only dead components). By Step 5, removals of dead components never free a dead component; so as the capture reduction is applied repeatedly at turn n, every not-yet-removed dead component stays trapped on the current board, provided all removals so far are of dead components. We show by induction that they are. Let the current application capture C. If C is old, C is dead by Step 3. If C is fresh, only condition 3a can fire, and no adjacent enemy stone is trapped on the current board. Suppose C were live. By Step 4 it has a dead enemy donor T adjacent. If T is already removed, its spots are empty and C is not trapped on the current board. If T is not yet removed, it is still trapped, and condition 3a fails for C. Either way C is not captured: a contradiction. So C is dead.

Step 7 (progress, and at most two removing applications). Consider the first application. Let T be a dead old component, white say. Its adjacent enemy stones trapped on the board lie in live components – dead neighbors do not exist by Step 5 – and live trapped components are fresh by Step 4. So condition 3b holds for T: no stone of T is of the turn, and every adjacent trapped component contains a stone of the turn that is not a q-stone. Every dead old component is captured on the first application. Now let T be a dead fresh component, through m_1 say. Its possible blocker for condition 3a is an adjacent trapped enemy component; by Step 5 it is live, by Step 4 fresh: the component Z of m_0. Suppose Z is adjacent to T and trapped on the current board. By Step 4, Z has a dead white donor adjacent. The donor is not T: were it, run the order of m_0 first – the whites removed at the resolution of m_0 are dead, none is adjacent to Z except possibly T, whose old part holds the liberty m_1 and whose stone m_1 is not yet placed; so either Z is libertyless there and suicides, or its one liberty is m_1, filled at the next placement whose enemy captures then remove Z; in both cases Z is dead, a contradiction. So the donor is a dead old white component. It is captured on the first application, freeing Z; on the second application Z is no longer trapped, condition 3a holds for T, and T is captured. So all dead components are removed in at most two removing applications, and the repetition terminates having removed exactly the dead components.

Step 8 (later stages are quiet, and conclusion). The removals at turn n only emptied intersections. An old component not adjacent to a fresh stone keeps a liberty of $\hat{a}$. A component with a fresh stone in or adjacent to it is not trapped on any turn $t < n$. A component adjacent to a removed stone has a liberty. So Cap_t removes nothing for $t < n$, and the recursion stops. The stones standing are the live components: by Steps 1–4 these are exactly the stones of $\hat{a}'$, and no recoloring occurred, there being no q-stone. So $OR(D^+)$ is the diagram of $\hat{a}'$ with stamps forgotten. If a move is a pass, the same argument runs with one fresh point. □

Remark 5.4. *Step 7 shows the repetition needs at most two removing applications on any board: there are at most two fresh components, and the second application*

8	.	n	b	q	k	b	n	r
7	.	p	p	p	p	p	p	p
6	.	r	.	.	.	.	.	.
5	p	.	.	.	.	.	.	.
4	P	.	.	.	.	.	.	.
3	.	R	.	.	.	.	.	.
2	.	P	P	P	P	P	P	P
1	.	N	B	Q	K	B	N	R
	a	b	c	d	e	f	g	h

FIGURE 1. The state a^*, reached by the mirror pairs $a4/a5$; $Ra3/Ra6$; $Rb3/Rb6$. White pieces are in capitals. The conflicting simultaneous moves are the rook moves $m_0 : b3 \to b5$ and $m_1 = \mathcal{A}(m_0) : b6 \to b4$.

serves only a dead fresh component shielded by the other. This is consistent with the recursion depth statistic of Section 2.1. The lemma is also computer verified: exhaustively on the 3×3 *board, and on* 1.75 *million sampled commuting transitions at sizes up to* 19×19.

5.2. **Chess and faithfulness.** We now come to the main reason for studying faithfulness: it gives rise to a qualitative mathematical difference between chess and *Go*.

Theorem 5.5. $\widetilde{\mathit{chess}}$ *has no simple simultaneization that is faithful within distance one of classical play: equivalently, none of positive faithfulness radius. In particular, the universal simultaneization* $\mathrm{Sim}(\widetilde{\mathit{chess}})$ *is not simple.*

Proof. First a technical lemma.

Lemma 5.6. *Let* G_1 *be a simultaneization of* G_0*, and suppose that* $s \in B(G_1)$ *is obtained from the initial state by a game history consisting of* mirror pairs*, that is, of simultaneous pairs of the form* $(x_i, \mathcal{A}(x_i))$*. Then* $\mathcal{R}(s) = s$*, and consequently* $\mathcal{A}(I(s)) = I(s)$.

Proof. The mirror of the pair $(x, \mathcal{A}(x))$ is the pair $(\mathcal{A}(\mathcal{A}(x)), \mathcal{A}(x)) = (x, \mathcal{A}(x))$ itself. So applying the symmetry property of Definition 4.2, and the order invariance of E (the last property of Definition 4.1), along the game history, together with $\mathcal{R}(q_0) = q_0$, we get $\mathcal{R}(s) = s$. Then for any m, $E_{G_1}(s, m)$ is defined if and only if $E_{G_1}(\mathcal{R}(s), \mathcal{A}(m)) = E_{G_1}(s, \mathcal{A}(m))$ is defined, giving $\mathcal{A}(I(s)) = I(s)$. □

Now return to the proof of the theorem. Let G_1 be any simultaneization of $\widetilde{\mathit{chess}}$ faithful within distance one of classical play, with P_0 the White player, and consider the game history of G_1 consisting of the mirror pairs

$$a4/a5; \quad Ra3/Ra6; \quad Rb3/Rb6,$$

in standard chess algebraic notation. The moves of each pair do not interact, so both serializations are pseudo legal with the same outcome, and each state along this history is semi-classical. By the semi-classicality condition of Definition 4.16 together with faithfulness at the semi-classical states, applied inductively along the history, the history is legal in G_1 and for the resulting state $s^* \in B(G_1)$ we have

$\rho(s^*) = \{a^*\}$, with a^* the classical state of Figure 1. Note that this pins down $\rho(s^*)$ for every simultaneization G_1, whatever its branch semantics.

Now let m_0 be the White rook move $b3 \to b5$, and $m_1 = \mathcal{A}(m_0)$ the Black rook move $b6 \to b4$. Each order of these moves blocks the other: after $b3 \to b5$ the Black rook's path is blocked at $b5$, while after $b6 \to b4$ the White rook's path is blocked at $b4$. So in $\mathrm{Sim}(\widetilde{chess})$ this pair resolves by the lowest priority case of the serialization map P, and

$$P(a^*, m_0, m_1) = \{a_0, a_1\},$$

where in a_0 the White rook stands on $b5$ with the Black rook on $b6$, and a_1 is the mirror image of a_0. By faithfulness at the semi-classical s^*, the pair (m_0, m_1), being available in $\mathrm{Sim}(\widetilde{chess})$ at $\{a^*\}$, is available at s^*; so $s^*_1 = E_{G_1}(s^*, m_0, m_1)$ is defined, and by condition 3 together with faithfulness at s^*_1 – a state one turn from s^* – $\rho(s^*_1)$ is a non-empty subset of $\{a_0, a_1\}$. Moreover s^*_1 is not final: it is within distance one of the semi-classical s^*, so the finality clause of faithfulness applies, and neither a_0 nor a_1 is final in $\widetilde{chess}$.

Suppose that $\rho(s^*_1)$ is a singleton, say $\{a_0\}$. The history of s^*_1 consists of mirror pairs, so by Lemma 5.6 $I(s^*_1)$ is $\mathcal{A}$ invariant. But by faithfulness at s^*_1, $I(s^*_1) = \mathbb{I}(a_0)$, which is not $\mathcal{A}$ invariant: in a_0 White has rook moves from $b5$, while Black has no rook moves from $b4$. The case of $\{a_1\}$ is mirrored. Thus a singleton value is forbidden by the symmetry of G_1, and $\rho(s^*_1) = \{a_0, a_1\}$.

By faithfulness at s^*_1, $I(s^*_1) = \mathbb{I}(a_0) \cap \mathbb{I}(a_1)$, which is computed directly: for each player it consists of the pass move, the moves of the c through h pawns, and the four knight moves. There are no queen side rook moves, as these rooks stand on different squares in a_0 and a_1, and no a or b pawn moves. Now suppose that G_1 is simple, so that there is a legally obtainable state c of $\widetilde{chess}$ with $\mathbb{I}(c)$ equal to this set. The interface $\mathbb{I}(c)$ forces every piece of c other than the two queen side rooks, the two a pawns and the two b pawns to stand on its starting square. One now checks the finitely many placements, or absences by capture, of the six remaining pieces, and each fails. For example, if the White b pawn is at $b2$, then $b3$ must be occupied, necessarily by a queen side rook or the Black b pawn, giving respectively a legal rook move from $b3$ and the legal capture $b3 \times c2$, neither of which is in the set; while if the White b pawn is absent, having been captured, and $b2$ is unoccupied, then the $c1$ bishop has the legal move to $b2$, again not in the set. So we have a contradiction, and G_1 is not simple. □

5.3. **Mirror-tie chess.** The faithfulness hypothesis in Theorem 5.5 cannot be dropped. Define the *mirror-tie game* G_1: play proceeds as in $\widetilde{chess}$ but with simultaneously declared moves, as long as the two moves of each turn do not genuinely interact – both orders of play legal and with the same outcome. Thus each state s so reached is semi-classical, with $\rho(s)$ the singleton of its classical state. At the first genuinely conflicting move pair the game locks: only the pass moves remain available. After two consecutive passes the game ends in a tie.

The conditions of Definition 4.16 are readily verified; the locked states, having only passes available, are exempt from the simplicity condition, and at every other state the interface is that of a classical state. So G_1 is a simple simultaneization of $\widetilde{chess}$, of faithfulness radius zero. (Fully degenerate examples also exist: the game in which only passes are ever available is a simple simultaneization of any G_0

with a pass move, vacuously.) Together with Theorem 5.5: the radius spectrum of $\widetilde{chess}$ is $\{0\}$ – its conflicts can be refused, but not played out – while the radius spectrum of *Go* is all of $\mathbb{N} \sqcup \{\infty\}$, by Theorem 5.2. Nothing in the mirror-tie construction is special to chess, so every game of our standing kind has zero in its radius spectrum. Within the hierarchy of *Go*: *SGo* continues past every conflict at radius zero, embracing the divergences; each turn of delay buys one level; *DSGo*, all delay, is faithful everywhere. The pattern of the objective reduction – prune the state, against the branch evolution, by the game's own rules – is not special to *Go*; we develop it only here.

Question 5.7. *Which subsets of* $\mathbb{N} \sqcup \{\infty\}$ *arise as radius spectra? Chess and Go realize* $\{0\}$ *and the whole of* $\mathbb{N} \sqcup \{\infty\}$. *We do not know whether a spectrum must be an initial segment. Nor do we know a game of finite positive absolute radius – a spectrum with finite positive supremum – but it is hard to imagine an obstruction.*

How does the radius spectrum sit among known invariants of games? The classical invariants are functionals of the scoring. The Conway value, the temperature, the outcome class: each is computed from the terminal values, by backward induction [4, 2]. The same holds for the invariants obtained by perturbing the turn mechanism. Under auction play, the Richman value is a threshold for winning [16, 5]. Under random turn order, the value is a winning probability [22]. The complexity class of a game family is the cost of deciding the winner; notably it does not draw our line, as generalized chess and *Go* with a simple Ko rule are both EXPTIME complete [8, 26]. The radius spectrum is of the same species – an invariant born of perturbing the turn mechanism – but it alone is blind to the scoring. It is a function of the dynamics alone (Lemma 4.20): replacing the scoring of *Go* by any other function of final displays changes nothing in Theorem 5.2. Purely dynamical equivalences of automata are the subject of concurrency theory: bisimulation, and the trace theory of commuting actions [18, 17]. A commuting turn is a diamond in that language, and $\mathrm{Sim}(G_0)$ resembles the passage from interleaving semantics to true concurrency. We do not know a numerical invariant of that theory in the role of the radius.

6. Informed play and concluding remarks

The focus of this section is our informed play experiments: their methodology and partial results, for both games. All figures below are for the 5×5 board, and may be compared directly with the random play columns of the table of Section 2.1. The restriction to 5×5 is due to computational cost: for *SGo*, estimating the equilibrium with our very blunt methods is expensive even on 5×5.

6.1. **Equilibrium guided play.** Our equilibrium guided play is computed as follows: at each turn both players are offered a random menu of five moves; the resulting 5×5 matrix of joint outcomes is filled by averaging the final margins of short uniformly random continuations;[8] and each player samples their move from an optimal mixed strategy of this zero sum matrix game. Three technical findings. First, at about 74% of the turns the estimated one turn game has *no pure saddle*

[8]Rollouts use the standard Monte Carlo playout convention: uniformly random moves, except that a player does not fill their own single point eyes and does not play suicide, passing when no other move is available.

point, so the players must randomize – consistent with Section 4.2, where mixing is proved necessary already on the 3×3 board (re-estimating the matrices at fivefold rollout depth, on fixed states and menus, lowers the mixing fraction from 74% to 54% on a control sample; genuine mixing remains). Second, the within turn value spread is large, about 16 points of the twenty-five: the joint move choice of a single turn swings most of the board. Third, the summary table of the next subsection shows that equilibrium guided play leaves the creation of entanglement essentially unchanged relative to the random play columns of the table of Section 2.1 (3.2 q-stones created per game against 3.3): the entanglement is a stable feature of the game, of order one under random and informed play alike.

6.2. **Versus informed play in classical *Go*.** For the classical game we use greedy rollout guided play: at each move the player evaluates up to six candidate moves, and the pass, by twelve random rollouts each (the convention of the preceding footnote), and plays the best. The summary of informed self play, for both games, in the format of the table of Section 2.1 (162 and 165 games; unbounded play in both cases; these runs predate the depth statistic and report branch counts instead):

	SGo	classical *Go*
mean margin	11.7	11.4
draws	5.6%	6.7%
games ending on their own	100%	100%
mean length in turns	21.2	17.8
q-stones created per game	3.2	–
q-stones on the final display	2.3	–
peak number of branches	4.2	–
final number of branches	3.2	–

For *SGo* the table is close to its random play columns: the margin eases (11.7 against 13.6), the draw rate and game length barely move, and the entanglement is unchanged in creation and slightly more resolved by the end. For the classical game the change is termination: informed play supplies the endings that random play never reaches – every game now ends on its own, in 17.8 turns on average – while *SGo* terminates by rule either way.

At this shallow search depth no measurable first mover advantage emerges in *Go* (signed mean margins statistically zero). Hence we are not correcting by any Komi, which is also why draws appear for *Go* in the table above. We note that *Go* is solved on the 5×5 board – a win for Black by 25 points under standard rule formalizations [32] – a reminder that these are partial, shallow search depth results.

Whether the above observations persist for equilibrium guided play on large boards, along with the sharpness observed in Section 2.1, we leave to future study.

6.3. **An Elo ladder.** Sharp dynamics alone do not distinguish a deep game from a coin flip. A standard proxy for depth is the length of the improvement ladder: how much playing strength a fixed family of agents can climb before it saturates [15]; strength against computational budget is likewise standard in the study of game-playing programs [14]. We ran one ladder for both games, on the 5×5 board, with the same architecture: Monte Carlo agents indexed by a rollout budget, choosing from a random menu of five moves by random playouts, the opponent modeled as random. Adjacent budgets played 100 games each, colors alternating; the final pair played 60. The Elo gap of a pair is $400\log_{10}(s/(1-s))$, with s the stronger agent's mean score.

budgets	*SGo*	*Go*
0 v. 16	+308	+173
16 v. 128	+330	+89
128 v. 1024	+215	+63
1024 v. 2048	+52	−45
2048 v. 4096	0	–
ladder	≈ 900	≈ 300

Both ladders saturate. That of *Go* does so by budget 1024: its last rung is negative, within noise of zero. That of *SGo* climbs about three times higher, and saturates about sixteen times later – a reflection of the fact that Elo is an exponential type scale. Direct matches between non-adjacent rungs agree with the summed gaps, in both games: the ladders hide no cycles. A control with komi one half, which eliminates draws, leaves the ladder of *Go* essentially unchanged: the disparity is not an artifact of the draw channel. These figures must be properly interpreted. They are relative to this agent family, and they lower bound the depth rather than measure it. Still, on the same board and by the same yardstick, the ladder of *SGo* is much taller.

6.4. **Lazy evaluation.** Finally, we note that the pattern of resolution in *SGo* – branchings accumulate silently until objective information forces their evaluation – is also familiar in computer science, notably in lazy evaluation [10, 13], where a deferred computation is forced exactly when its value is demanded.

References

[1] R. Alur, T. A. Henzinger, and O. Kupferman, *Alternating-time temporal logic*, Journal of the ACM, 49 (2002), pp. 672–713. 4

[2] E. R. Berlekamp and D. Wolfe, *Mathematical Go: Chilling Gets the Last Point*, AK Peters, Natick, MA, 1994. 1, 5.3

[3] A. Cincotti and H. Iida, *The game of synchronized cutcake*, in IEEE Symposium on Computational Intelligence and Games (CIG 2007), IEEE, 2007, pp. 374–379. 1

[4] J. H. Conway, *On Numbers and Games*, Academic Press, 1976. 1, 4.3, 5.3

[5] M. Develin and S. Payne, *Discrete bidding games*, Electronic Journal of Combinatorics, 17 (2010), p. R85. 5.3

[6] L. Doyen and J.-F. Raskin, *Games with imperfect information: theory and algorithms*, in Lectures in Game Theory for Computer Scientists, Cambridge University Press, 2011. 4.12

[7] J. Eisert, M. Wilkens, and M. Lewenstein, *Quantum games and quantum strategies*, Physical Review Letters, 83 (1999), pp. 3077–3080. 1

[8] A. S. Fraenkel and D. Lichtenstein, *Computing a perfect strategy for $n\times n$ chess requires time exponential in n*, Journal of Combinatorial Theory, Series A, 31 (1981), pp. 199–214. 5.3

[9] D. Gale, H. W. Kuhn, and A. W. Tucker, *On symmetric games*, in Contributions to the Theory of Games, Vol. I, vol. 24 of Annals of Mathematics Studies, Princeton University Press, 1950. 4.2

[10] P. Henderson and J. H. Morris, Jr., *A lazy evaluator*, in Proceedings of the 3rd ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL '76), ACM, 1976, pp. 95–103. 6.4

[11] M. Huggan, R. J. Nowakowski, and P. Ottaway, *Simultaneous combinatorial game theory*. arXiv:1810.02870, 2018. 1, 4

[12] M. A. Huggan, *Studies in Alternating and Simultaneous Combinatorial Game Theory*, ph.d. thesis, Dalhousie University, Halifax, Nova Scotia, Canada, 2019. 1

[13] J. Hughes, *Why functional programming matters*, The Computer Journal, 32 (1989), pp. 98–107. 6.4
[14] A. L. Jones, *Scaling scaling laws with board games*. arXiv:2104.03113, 2021. 6.3
[15] F. Lantz, A. Isaksen, A. Jaffe, A. Nealen, and J. Togelius, *Depth in strategic games*, in Workshops of the Thirty-First AAAI Conference on Artificial Intelligence: What's Next for AI in Games?, 2017. 6.3
[16] A. J. Lazarus, D. E. Loeb, J. G. Propp, W. R. Stromquist, and D. H. Ullman, *Combinatorial games under auction play*, Games and Economic Behavior, 27 (1999), pp. 229–264. 5.3
[17] A. Mazurkiewicz, *Trace theory*, in Petri Nets: Applications and Relationships to Other Models of Concurrency, Advances in Petri Nets 1986, vol. 255 of Lecture Notes in Computer Science, Springer, 1987. 5.3
[18] R. Milner, *Communication and Concurrency*, Prentice Hall, 1989. 5.3
[19] J. F. j. Nash, *Equilibrium points in n-person games*, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49. 1, 4.2
[20] M. J. Osborne and A. Rubinstein, *A Course in Game Theory*, MIT Press, 1994. 4.2
[21] R. Penrose, *The Emperor's New Mind*, Oxford University Press, 1989. 1
[22] Y. Peres, O. Schramm, S. Sheffield, and D. B. Wilson, *Random-turn hex and other selection games*, American Mathematical Monthly, 114 (2007), pp. 373–387. 5.3
[23] M. O. Rabin and D. Scott, *Finite automata and their decision problems*, IBM Journal of Research and Development, 3 (1959), pp. 114–125. 4.12
[24] A. Ranchin, *Quantum Go*. arXiv:1603.04751, 2016. 1
[25] J. H. Reif, *The complexity of two-player games of incomplete information*, Journal of Computer and System Sciences, 29 (1984), pp. 274–301. 4.12
[26] J. M. Robson, *The complexity of go*, in Information Processing 83, Proceedings of the IFIP 9th World Computer Congress, 1983, pp. 413–417. 1, 5.3
[27] U. Schwalbe and P. Walker, *Zermelo and the early history of game theory*, Games and Economic Behavior, 34 (2001), pp. 123–137. 4.2
[28] L. S. Shapley, *Stochastic games*, Proceedings of the National Academy of Sciences, 39 (1953), pp. 1095–1100. 4
[29] M. Thielscher, *The general game playing description language is universal*, in Proceedings of the 22nd International Joint Conference on Artificial Intelligence (IJCAI 2011), AAAI Press, 2011. 4
[30] J. Tromp and G. Farnebäck, *Combinatorics of Go*, in Computers and Games (CG 2006), vol. 4630 of Lecture Notes in Computer Science, Springer, 2007, pp. 84–99. 5.1
[31] J. Tromp and B. Taylor, *The logical rules of Go*. https://tromp.github.io/go.html, 1995. 5.1
[32] E. C. D. van der Werf and M. H. M. Winands, *Solving Go for rectangular boards*, ICGA Journal, 30 (2009), pp. 77–88. 6.2